

PRODUCT OVERVIEW · V1

PantryPal

Replace the grocery phone call. Two people, one shared pantry. Wife marks items low in 2 taps. Husband sees the live list at the store.

Expo SDK 56

React Native

TypeScript

Supabase

Zustand

Expo Router

expo-notifications

expo-location

expo-camera

HOW IT WORKS

The practical flow



Wife notices milk is low
Opens app



Taps Milk → Mark Low
2 taps total



Husband gets push alert
Instant, background



Husband drives to store
GPS detects arrival



Grocery list auto-opens
Geofence trigger



Taps "Got it"
Disappears on both phones

No
The

FEATURES

What's built

LIVE

LIVE



Pantry Screen

- 50 pre-seeded items — Fridge, Pantry, Freezer sections
- Tap any item → marks LOW with amber highlight + badge
- Add custom items, search and filter
- Emoji per item, category grouping

LIVE



Stores

- Add grocery stores with GPS coordinates
- Assign preferred store per pantry item
- Geofence triggers when you physically arrive
- 3-min debounce prevents GPS bounce re-triggers

LIVE



Barcode Scanner

- Scans EAN-13, EAN-8, UPC-A, UPC-E, Code128
- Looks up product name via Open Food Facts API
- Pre-fills the add-item form automatically
- Haptic feedback on successful scan

LIVE



Push Notifications



Grocery List

- Live list of every item marked LOW
- Real-time sync — appears on spouse's phone in <2 seconds
- Shopping mode keeps screen on with larger text
- Tap row → "Got it" removes from both phones instantly

LIVE



Receipt Scanning

- Scan or upload grocery receipts
- Supabase Edge Function parses with AI
- Scanned items auto-populate pantry
- History of past receipts preserved

LIVE



Household

- Two people share one household account
- Invite spouse via deep-link code (pantrypal://join)
- All data is household-scoped with Supabase RLS
- No user can see another household's data

LIVE



Offline Mode

- Sent when any item is marked LOW
- Supabase Edge Function + Expo Push Tokens
- Works in background and when app is closed
- Low-stock alerts for expiring items
- All writes queue locally when no connection
- Sync status pill: Synced / Queued / Reconnecting
- Grocery list stays readable at the store offline
- Auto-flushes queue every 8s when back online

ARCHITECTURE

How data flows

households	One per family unit. Holds the invite_code for deep-link joining. All other data hangs off this.
household_members	Maps users to households. Stores each member's push_token — updated on every app launch in case it changes on reinstall.
items	The master item list. Key flag: is_low boolean. Tracks who marked it low and who got it. Indexed for fast household queries.
stores	Grocery stores with lat/lng for geofencing. Items can have a preferred_store_id to route shopping by aisle.
Realtime	Supabase Realtime subscription on the items table. Changes appear on both phones in under 2 seconds. Re-subscribes on app foreground.
Edge Functions	parse-receipt (AI receipt parsing), notify-low-item (push on mark low), notify-store-arrival (geofence push). All webhook-secured.
RLS	Row Level Security enforced on every table. Users can only see data from their own household. No app-level guard needed — the database enforces it.

RESILIENCE

Edge cases handled

**Both mark same item simultaneously**

is_low=true is idempotent. Last write wins, no data loss, no conflict.

**No internet at the store**

Grocery list cached in Zustand. Reads work offline. Writes queue and show "Queued" badge.

**Realtime subscription drops**

AppState listener re-subscribes on foreground. "Reconnecting..." banner after 5s without connection.

**New phone / reinstall**

Push token upserted on every app launch. Household membership persists in Supabase.

**GPS bounce in parking lot**

3-minute debounce on geofence re-entries. Same store won't re-trigger within the window.

**Two people join with same invite code**

DB unique constraint on household_members prevents duplicates. Atomic race-safe check.